

SMART TRACK SPM

Chapter 1: Rational Numbers

1. Integers

Integers are numbers that can be positive, negative, or zero. They are used to represent real-world situations like temperature and money.

- 1 +5 → Positive number (above zero)
- 2 0 → Neutral (no gain, no loss)
- 3 -3 → Negative number (below zero)

Example: If temperature drops from 3°C to -2°C, it means it goes below freezing.

■ **Tip:** Integers do NOT include fractions or decimals.

2. Number Line

A number line helps you visualize numbers and compare them easily.

← -3 -2 -1 0 1 2 3 →

- 1 Numbers on the RIGHT are bigger
- 2 Numbers on the LEFT are smaller

Example: 2 is greater than -1 because it is on the right.

■ **Tip:** Always imagine movement on the number line when solving problems.

3. Addition & Subtraction

Think of addition and subtraction as movement.

- 1 + means move RIGHT
- 2 - means move LEFT

- 1 $5 + 3 = 8$
- 2 $5 + (-2) = 3$
- 3 $-4 + 6 = 2$

Special Rule: $5 - (-3) = 5 + 3 = 8$

■ **Tip:** Two negatives together become positive.

4. Multiplication & Division

Operation	Result
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(+)(+)	+
(+)(-)	-
(-)(+)	-
(-)(-)	+

1 $-5 \times (-4) = 20$

2 $10 \div (-2) = -5$

■ **Tip:** Same signs = positive, different signs = negative.

5. Fractions & Decimals

Fractions and decimals can also be positive or negative.

1 $1/2 \rightarrow$ Positive

2 $-3/4 \rightarrow$ Negative

3 $0.5 \rightarrow$ Positive

4 $-2.3 \rightarrow$ Negative

■ **Tip:** The sign affects the whole number, not just part of it.

6. Real-Life Applications

Rational numbers are used in everyday life.

1 Temperature: $+30^{\circ}\text{C}$ or -5°C

2 Money: Profit (+), Loss (-)

3 Bank balance: -200 means debt

4 Height: above or below sea level

■ **Tip:** Understanding rational numbers helps you solve real-world problems.